GG-M-00112b(DSA-DM)
12 April 1965
Interim Revision of
Fed. Spec. GG-M-112a
15 May 1951

INTERIM FEDERAL SPECIFICATION

MANOMETER OUTFIT, SPINAL-FLUID (WATER-TYPE)

This interim Federal Specification was developed by the Defense Medical Supply Center, Third Avenue and 29th Street, Brooklyn, New York, 11232, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above. The General Services Administration has authorized the use of this interim Federal specification as a valid exception to Federal Specification GG-M-112a, dated 15 May 1951.

1. SCOPE

1.1 The specification covers one type of spinal-fluid, water-type manometer, complete with 3-way stopcock, 3-1/2 inch, 20-gage spinal needle, and sterilizing tray.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification, to the extent specified herein:

FSC 6515

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Federal Specifications:

GG-I-526	Instruments, Dental and Surgical; General Specifications.
GG-N-196	Needles, Hypodermic, For Luer Syringes.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.

Federal Standard:

Fed. Std. No.66 Steel: Chemical Composition and Hardenability.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards, and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington 25, D.C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards, and the Index of Federal Specifications and Standards, from established distribution points in their agencies.)

Military Specifications:

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

Military Standards:

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

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(Copies of military specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN STANDARDS ASSOCIATION, INC.

ASA Z70.1-1955 (UDC615.47) American Standard Dimensions of Glass and Metal Luer Tapers for Medical Applications.

(Application for copies should be addressed to the American Standards Association, Inc., 10 East 40th Street, New York, N.Y. 10016).

COPPER DEVELOPMENT ASSOCIATION, INC.
Standard Handbook For Copper and Copper Alloys.

(Application for copies should be addressed to Copper Development Association, Inc., 405 Lexington Avenue, New York, N.Y. 10017)

3. REQUIREMENTS

3.1 Material

3.1.1 Glass. Glass parts shall be fabricated of clear flint glass or of borosilicate glass, suitable for the intended purpose and meeting the requirements herinafter specified. There shall be no ring strain present in the fabricated parts, and any longitudinal strain present shall be faint and highly diffused.

3.1.2 Brass. Brass components shall be finished as specified in 3.11. Unlesss otherwise specified herein, brass components shall be fabricated from the following copper alloys:

Copper Alloy No. 360 (Free Cutting Brass) - Three-way stopcock.

Copper Alloy No. 260 (Cartridge Brass, 70%) - Sterilizing tray.

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3.1.3 Stainless steel. The stainless steel shall conform to AISI 304, of Standard No.66 stainless steel; shall comply with the corrosion test requirements of Class 2, Composition G of GG-I-526 and hardness shall be B75 minimum."

3.1.3.1 Needle. The needle shall conform to the applicable requirements of GG-N-196 for type I needles. As an alternate, the cannula may be fabricated from welded and drawn tubing and the material for cannula, whether fabricated from seamless drawn or welded and drawn tubing, may be type 304 AISI stainless steel.

3.2 Design. The spinal fluid manometer shall be of the water-type, consisting essentially of two capillary tubes, a spinal needle, a three-way stopcock, and a metal sterilizing tray. Standard Luer tips shall be tapered in accordance with figure 1 of ASA Z70.1-1955, and shall meet the requirements as specified in 3.2.1. The standard female Luer lock connectors (flanges and hub cavities) shall be in accordance with figure 3 of ASA standard and shall meet requirements as specified in 3.2.2. The standard male Luer lock connectors shall be in accordance with figure 4 of the ASA standard and shall meet requirements as specified in 3.2.3.

3.2.1 Luer slip tapered tips. The components having Luer slip tapered tips shall make secure, watertight unions with needle hubs having a corresponding taper. The tip shall be tested for proper taper and dimensions as specified in 4.3.5. When so tested, tips shall fit the ring gage within the "GO"- "NO GO" limits of the gage as shown on figure 7 of the ASA standard. Tips that do not fit the gage within these limits shall be considered not acceptable. In addition, when this ring gage is fitted onto the tip as specified above, there shall be no movement of the gage on the tip when the gage is pressed lightly from side to side in all directions. The tips shall also be tested for fluid tightness as specified in 4.3.8 and when so tested, shall not show leakage exceeding the formation of a falling drop in 10 seconds.

3.2.2 Female Luer connectors. The components having female Luer connectors shall make secure, watertight unions with male Luer slip connectors and Luer sock syringes, where applicable. The female Luer lock connector shall be tested for depth and diameter as specified in 4.3.6. The hub cavities shall fit the plug gage within the "GO" and "NO GO" limits of the feeler gage as shown on figures 5 and 6 of the ASA standard. Hubs which do not fit the plug gage within these limits shall be considered not acceptable. In addition, when the hub is fitted onto the plug gage, as specified, there shall be no movement of the hub of the component on the gage when component is pressed lightly from side to side in all directions. The female Luer connectors shall also be tested for fluid tightness as specified in 4.3.9 and, when so tested, shall not show leakage exceeding the formation of a falling drop in 10 seconds.

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3.2.3 Male Luer lock connectors. The component having male Luer lock connectors shall make secure, watertight union with needle hubs or needle adapters having corresponding taper, and shall be of such dimension and taper as to pass test as specified in 4.3.7. These tips shall fit the test gage within the "GO" - "NO GO" limits of the gage as shown on figure 7 of the ASA standard. Tips which do not fit the gage within these limits shall be considered not acceptable. In addition, when this ring gage is fitted onto the tip, as specified in 4.3.7, there shall be no movement of the gage on the tip when the gage is pressed lightly from side to side in all directions. The Luer lock tip shall also be tested with the thread checking gage as specified in 4.3.7. When fully seated, the thread checking gage shall not be capable of being further turned and shall show no movement when pressed lightly from side to side in all directions. Luer lock shall be tested for fluid-tightness as specified in 4.3.10. When so tested, there shall be no leakage exceeding one falling drop between the tip and the hub of the thread-checking gage in 10 seconds.

3.3 Capillary tubes. The tubes, one lower and one upper, shall be of glass. The inside diameter shall be 1.5 $\pm$ 0.25 millimeters and the outside diameter 7.0 $\pm$ 1.0 millimeters.

3.3.1 Lower tube. The lower tube shall be fitted at the lower end with a standard male Luer lock tip (connector), as specified in 3.2 and 3.2.3, that will make a nonleakable connection with the Luer cavities of the stopcock specified in 3.5. The upper end shall be fitted with an adapter for connection with the upper tube; the face end of the adapter shall have a standard Luer cavity as specified in 3.2 and 3.2.2, except that it shall not be provided with a flange.

3.3.2 Upper tube. The lower end of the upper tube shall be provided with an adapter for connection with the lower tube; the free end of the adapter shall have a standard Luer tip, as specified in 3.2 and 3.2.1, which shall make a non-leakable connection with the lower tube.

3.3.3 Graduation scale. The lower tube shall be graduated from 3 centimeters (30 mm.) to 30 centimeters (300 mm.), and the upper tube shall be graduated from 33 centimeters (330 mm.) to 60 centimeters (600 mm.). The scale on the glass tubes shall be marked in 2 millimeter graduations with main graduations every 1 centimeter (10 mm). The main graduations shall be not less than 3 millimeters in length. Figures indicating the number of centimeters shall appear either at each main graduation or at alternate main graduations. The subgraduations shall be plainly distinguished from the main graduations and shall be not less than 2 millimeters in length. The graduation markings shall be consecutive when the upper and lower tubes are joined. Connection of the two tubes may obliterate the glass tube graduations for not more than 5 centimeters. All markings shall be clearly legible and either fused in, or fused on the glass tubes.

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3.4 Spinal puncture needle. The needle shall be 20 gage, 3-1/2 inches long, and shall conform to requirements of Specification GG-N-196 for type I needles, style b, short bevel point, except hub shall be as specified in 3.2 and 3.2.2. The hub shall be so shaped that it will not roll and will provide a satisfactory finger grip for attaching the needle to a stopcock. The entire cavity wall down to the entrance of the lumen shall be well plated. The hub shall be slotted to receive the guide provided on the knob of the stylet. Each needle shall be fitted with a wire stylet. The wire shall be capable of easy insertion into, and withdrawal from, the needle. The wire shall be of adequate length to permit one end to be permanently secured into a knob and the opposite end to be ground flush with the cutting edge of the cannula (i.e., the beveled point of the cannula). The wire shall be of sufficient diameter so as to prevent such movement of the stylet which might permit the cut tip of the stylet to be bent while the stylet is in position in the needle, (i.e., bevel of the stylet tip shall blend with bevel of point of the cannula presenting a single uniform surface with point of cannula). The knob shall be of conventional design, having a taper tip that shall properly fit the taper slip of the needle hub, and a guide of adequate style and design to fit into the slotted edge of the needle hub and hold the ground edge of the wire stylet in proper position. The wire for stylet shall be corrosion-resistant steel containing 8.0 to 10.0 percent nickel, 18.0 to 20.0 percent chromium, and not more than 0.08 percent carbon, or AISI 304 stainless steel. Spinal puncture needle shall meet requirements as specified in 4.3.12.

3.5 Three-way stopcock. The stopcock shall be of brass. It shall be capable of directing the flow of the spinal fluid from the spinal puncture needle either upward at a right angle into the manometer tube or horizontally into a syringe, test tube, etc., or of shutting off the flow. The stopcock shall have three connectors: A tip with a standard Luer taper, as specified in 3.2 and 3.2.1, which will make a nonleakable union with the spinal puncture needle; the other two openings shall be standard female Luer connectors, as specified in 3.2 and 3.2.2. The stopcock and first adapter may take the column fluid to the graduation at 3 centimeters. The valve directing the flow shall be readily and easily controlled in such a manner that the three operations of the stopcock are accomplished within one half-turn of the controlling handle. The stopcock shall pass the fluid tightness tests as specified in 4.3.11. No part of the stopcock nor any of the connection points of the stopcock shall show leakage exceeding the formation of a falling drop in 10 seconds.

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3.6 Sterilizing tray. The sterilizing tray shall be made of stainless steel or brass sheet, hard temper, chromium-plated or nickel-plated. The stainless steel or brass shall be of sufficient thickness and so constructed that it will not bend under ordinary use. The tray shall be fitted with permanently attached metal clasps, or spring clamps of the same composition as the tray, or other similar means to hold securely all parts of the outfit. A metal handle, 1 inch high, shall be attached at the center of the tray.

3.6.1 Dimensions. The sterilizing tray shall be approximately 13.5 inche long by 1.0 inch wide.

3.7 Resistance to boiling. The capillary tubes shall show no evidence of scumming, chipping, or cracking, resulting from the boil test specified in 4.3.3.

3.8 Permanency of marking. Capillary tubes shall be subjected to the permanency of marking test as specified in 4.3.4. Tubes shall be considered to pass the test if not more than one-half of any graduation line or numeral has been lost, and the area showing complete pigment loss is less than two percent of the total pigmented area, and if the legibility of any portion of the markings is not seriously impaired.

3.9 Solubility. No pink color shall develop in the testing solution when the tubes are tested as described in paragraph 4.3.1.

3.10 Resistance to thermal shock. The capillary tubes shall not break (due to temperature changes) when submitted to the thermal shock test as specified in 4.3.2.

3.11 Plating. The plated metal parts shall conform to the applicable requirements of Specification GG-I-526, except that a chromium coating is not required.

3.12 Identification marking. Unless otherwise specified, glass parts and stopcock shall be permanently marked with the name or registered trademark of the manufacturer.

3.13 Instruction sheet. A complete set of instructions for the use and care of the manometer outfit shall be supplied with each apparatus.

3.14 Workmanship. The workmanship shall be first class throughout. The manometer outfits shall be free from defects which detract from their appearance or which may impair their serviceability.

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4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105.

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as indicated in table below:

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Component	Characteristic	Rqmt.	Test procedure	Unit of product	Insp. level	AQL (percent) defective
Capillary tube	Boil test	3.7	4.3.3	1-tube	S-2	*
Capillary tube	Permanency of marking	3.8	4.3.4	1-tube	S-2	*
Capillary tube	Solubility	3.9	4.3.1	1-tube	S-2	*
Capillary tube	Thermal shock	3.10	4.3.2	1-tube	S-2	*
Needle	Requirements of GG-N-196	3.4	4.3.12	1-needle	S-2	*
Stainless steel parts	Requirements of GG-N-526	3.6	GG-I-526	1-part	S-2	*
Plated metal parts	Requirements of GG-I-526	3.11	4.3.13	1-part	S-2	*
Components with Luer slip tips	Taper	3.2.1	4.3.5	1-each	II	1.0
	Fluid tightness	3.2.1	4.3.8	1-each	S-4	1.5
Components with female Luer connectors	Dimensions and taper	3.2.2	4.3.6	1-each	II	1.0
	Fluid tightness	3.2.2	4.3.9	1-each	S-4	1.5
Components with male Luer lock connectors	Dimensions and taper	3.2.3	4.3.7	1-each	II	1.0
	Fluid tightness	3.2.3	4.3.10	1-each	S-4	1.5
Stopcock	Fluid tightness	3.5	4.3.11	1-stop-cock	S-4	1.5

* Acceptance number shall be zero (0) for all sample sizes.

4.2.3 Raw materials. A sufficient quantity of each lot of each metal used in the construction of the manometer shall be selected for chemical analysis.

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4.2.4 Components. Sufficient samples of the components shall be selected from each lot of material used in the manufacture of the manometers to perform tests as required by the applicable subsidiary specifications or by tests specified herein. When sample size for components covered by the subsidiary specifications is not specified, the sample size for each test shall be as indicated above. No test result shall fall below the minimum or exceed the maximum values specified. The sample unit for testing shall be the quantity necessary to perform all required tests one time each. Not less than three (3) samples shall be taken regardless of lot size.

4.2.5 Where possible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.3.1 Solubility. The glass capillary tube shall be washed and rinsed in distilled water at room temperature for 10 minutes. They shall then be partially filled to one-fourth of their capacity with CO_2- free distilled water containing 0.2 percent U.S.P. standard phenolphthalein test solution and autoclaved at 15 pounds pressure for 30 minutes at not less than 248° F.

4.3.2 Thermal shock test. The glass capillary tubes shall be placed in an oven which has been previously heated to 100° ± 1° C., and allowed to remain therein at this temperature for at least 30 minutes. They shall then be removed from the oven and immediately plunged into water at a temperature of 5° C.

4.3.3 Boil test. The tubes shall be completely immersed in distilled water in a closed vessel having a small steam vent, and boiled continuously for 6 hours, boiling distilled water added when necessary to compensate for evaporation. The tubes shall be suspended in the water so that they do not come in contact with the containing vessel. Examination shall be made for compliance with 3.7.

4.3.4 Permanency of marking test. The capillary tubes shall be completely immersed in an open beaker containing hydrochloric acid solution, as described below. Autoclave the beaker containing the hydrochloric acid and tubes at 15 p.s.i.g. and 250° Fahrenheit (F.) for two periods of 30 minutes each. The 30-minute period shall begin at the instant the temperature of the hydrochloric acid solution in the autoclave reaches 245° F., as measured by a thermocouple immersed in the liquid. At the end of the 30-minute period, remove the tubes, rinse with cold tap water, and dry by wiping with cellulose tissue. Repeat the autoclaving operation and drying process, and then examine for loss of pigmentation. The hydrochloric acid solution shall be as follows:

Hydrochloric acid, 35 - 38 percent, U.S.P. or American Chemical Society (ACS) quality--10.0 milliliters (ml.)

Distilled water, quantity sufficient to make (q.s.) 1000 cc.

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4.3.5 Taper of male Luer slip tip. The male Luer tip shall be tested with the ring gage shown on figure 7 of ASA standard. This gage shall be slipped onto the tip with very light pressure.

4.3.6 Diameter and depth of hub cavities of female Luer connectors. The hub cavity of each female Luer connector shall be tested on the male Luer taper gage in accordance with plug gage of figure 5 of ASA standard, (except that it shall be constructed with a suitable fluid path). The hub shall be slipped onto this gage with very light pressure. The fit of the hub on this gage shall be tested by means of the feeler gage in accordance with figure 5 of ASA standard.

4.3.7 Dimensions and taper of male Luer lock tip. Each Luer lock tip shall be tested with the ring gage as shown on figure 7 of ASA standard. This gage shall be slipped onto the tip with very light pressure. In addition, the Luer lock tip shall be tested with the thread checking gage shown on figure 8 of the ASA standard. (This gage shall be shown on figure 8 of the ASA standard and shall be constructed with a suitable fluid path throughout its length). The thread checking gage shall be fitted to the tip and shall be capable of entering the tip and being fully seated.

4.3.8 Fluid tightness for male Luer slip tips. A gage, as shown on figure 8 of the ASA standard (except that it shall be constructed with a suitable fluid path), shall be attached to the male Luer slip. The free end shall be plugged and the water under pressure of 45 pounds per square inch (p.s.i.) shall be let in through the gage for 10 seconds.

4.3.9 Fluid tightness for female Luer connectors. The female Luer hubs shall be attached by means of a plug gage of figure 5 of ASA standard (except that gage shall have a fluid path) to a water pressure source, having a gage, of 45 p.s.i. for 10 seconds with free end plugged.

4.3.10 Fluid tightness of male Luer lock tip. The thread checking gage, as shown on figure 8 of the ASA standard (and having a suitable fluid path throughout its length), shall be firmly fitted to and fully seated (locked) with the Luer lock tip. The gage shall be attached to a water source having a pressure gage of 45 p.s.i. for 10 seconds with free end plugged.

4.3.11 Fluid tightness for stopcock. A gage, as shown on figure 8 of ASA standard (except that gage shall have a fluid path), shall be fitted to the male Luer slip tip of the stopcock. With the gage and stopcock in horizontal position, a hub of the stopcock shall be attached by means of a plug gage as shown on figure 5 of ASA standard (except that gage shall have a fluid path) to a water pressure source having a pressure gage. The water shall be run through the stopcock so as to fill any cavity space and the lumen of the Luer slip and the gages. The lumen of the gages shall be sealed by a piece of solid rubber slipped on to point and a pressure of 45 p.s.i. applied for 10 seconds. This test shall be performed for both female Luer hubs of the stopcock and the male Luer slip, as well as the ground joint between the spigot and the stopcock body.

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4.3.12 Needles. Needles shall be tested to determine conformance to the requirements of GG-N-196, except that reference to resistance to breakage shall be deleted and the following test substituted:

The needle hub shall be clamped in a fixed position. A force shall be applied to the cannula at a point one inch from bub in order to bend the cannula in a clockwise direction through an angle of 25 degrees from the original neutral position. A force shall then be applied in a counterclockwise direction to bend the cannula past its neutral position and to the angle indicated above. The needle cannula shall then be returned to its original neutral position. The complete operation described above shall be repeated 20 times. The cannula shall not break when subjected to this test.

4.3.13 Plating. The plated metal parts shall be tested to determine conformance with the requirements of GG-I-526.

4.4 Examination. The manometers shall be examined to determine compliance with all requirements contained in this specification.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One (1) manometer, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table I shall be packaged in each unit and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table I. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	not required	24 units

5.1.2.1 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Unit package shall be a carton of appropriate size constructed in accordance with Federal Specification PPP-B-636. Carton shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of, or damage to, the contents. Closure shall be adequate to prevent spillage of contents under normal handling.

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5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States of America.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type III, class 1 plywood; PPP-B-621, table III; or PPP-B-636, type CF, class weather-resistant.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix, and waterproofing shall conform with 30.4, of PPP-B-636.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall conform to PPP-B-601, table I, using type III, class 1 plywood; or shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 2 use; PPP-B-621, table I; or PPP-B-636, type CF, class domestic and method I closure shall be utilized.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

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5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Selection of applicable levels of packaging and packing. (see 5.1 and 5.2).

6.2 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Federal government requirements.

Custodians:
Army. - MD
Navy. - MS
Air Force. - 03

Review activities:
Army. - MD
Navy. - MS
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Preparing activity:
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CAUTION

NOTICE TO BIDDERS/OFFERORS

DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

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DEFENSE MEDICAL SUPPLY CENTER
3RD AVENUE AND 29TH STREET
BROOKLYN, N.Y., 11232

POSTAGE FEES PAID
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DEFENSE SUPPLY AGENCY

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SPECIFICATION ANALYSIS SHEET

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Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION MANOMETER OUTFIT, SPINAL-FLUID (WATER-TYPE) dated 12 April 1965
FEDERAL SPECIFICATION GG-M-00112b(DSA-DM)

ORGANIZATION (Of submitter)	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity)	DATE

DD FORM 1426. SND PPSO 33503

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